QPCode……

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP BCA (HONOURS) EXAMINATION OCTOBER 2026

FIFTH SEMESTER

DSE-MG5DSEBCA303-NEURAL NETWORKS

(2024 ADMISSION ONWARDS)

Duration: 2 hours Maximum Marks: 70

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries 2 marks

1. What is the purpose of the ReLU activation function? [K] [CO1]

2. What is a feature map in a Convolutional Neural Network? [K] [CO2]

3. What is the exploration–exploitation trade-off in reinforcement learning? [U] [CO3]

4. What is the main purpose of dimensionality reduction? [U] [CO4]

5. What is Unsupervised Learning? [K] [CO4]

[2 × 5 = 10]

Part B

Short Answer Questions

Answer any 5 questions

Each question carries 6 marks

6. Explain the backpropagation algorithm used for training a neural network. [U] [CO1]

7. Explain convolution operation, filters and feature extraction in CNNs. [U] [CO2]

8. Explain stride, padding and output dimension calculation in a convolutional layer. [U] [CO2]

9. Explain the architecture and working of a basic Recurrent Neural Network. [U] [CO3]

10. Explain the PCA algorithm and discuss how dimensionality reduction is achieved using PCA. [An] [CO4]

11. Explain the architecture of Deep Convolutional GAN. [U] [CO4].

12. Explain K-Means clustering as unsupervised learning technique. [U] [CO4]

[6 × 5 = 30]

Part C

Essay Questions

Answer any 2 questions

Each question carries 15 marks

13. Explain the fundamentals of neural networks in detail. Discuss feedforward computation, weights and biases. Also describe the basic features and uses of TensorFlow and PyTorch. [U] [CO1]

14. Explain Convolutional Neural Networks in detail. Discuss the design features of LeNet, AlexNet, VGG and ResNet. [K] [CO2]

15. Compare advanced RNN architectures-LSTM and GRU [An] [CO3]

[15 × 2 = 30]